

Supplementary Material for: 'Investigating Perceived and Modeled Transition Regions Between Acoustic Zones Around an Obstacle'*

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0 Acoustic Parameter Variation

The present section includes visualizations depicting the acoustic parameter variation along the lines. For example, in Fig. 1 and Fig. 2, the parameter value variation for C_{80} and EDT can be observed, showcasing a similar behavior as for T_S . In Fig. 3 the variation of T_S across three subbands can be observed for S_2 .

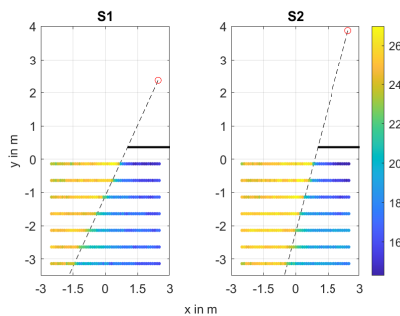


Fig. 1: C_{80} variation for S_1 and S_2 along the lines. The dotted lines indicate the geometric boundary.

1 Subject-related Metrics

The following section presents subject-related metrics extracted from the tracking data. The task completion time

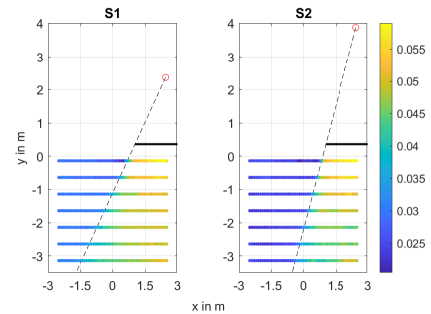


Fig. 2: EDT variation for S_1 and S_2 along the lines. The dotted lines indicate the geometric boundary.

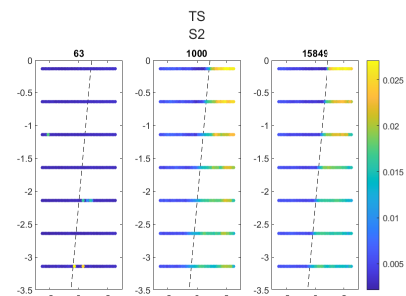


Fig. 3: Subband variation of T_S for S_2 for 63 Hz, 1000 Hz and 15849 Hz. The dotted lines indicate the geometric boundary.

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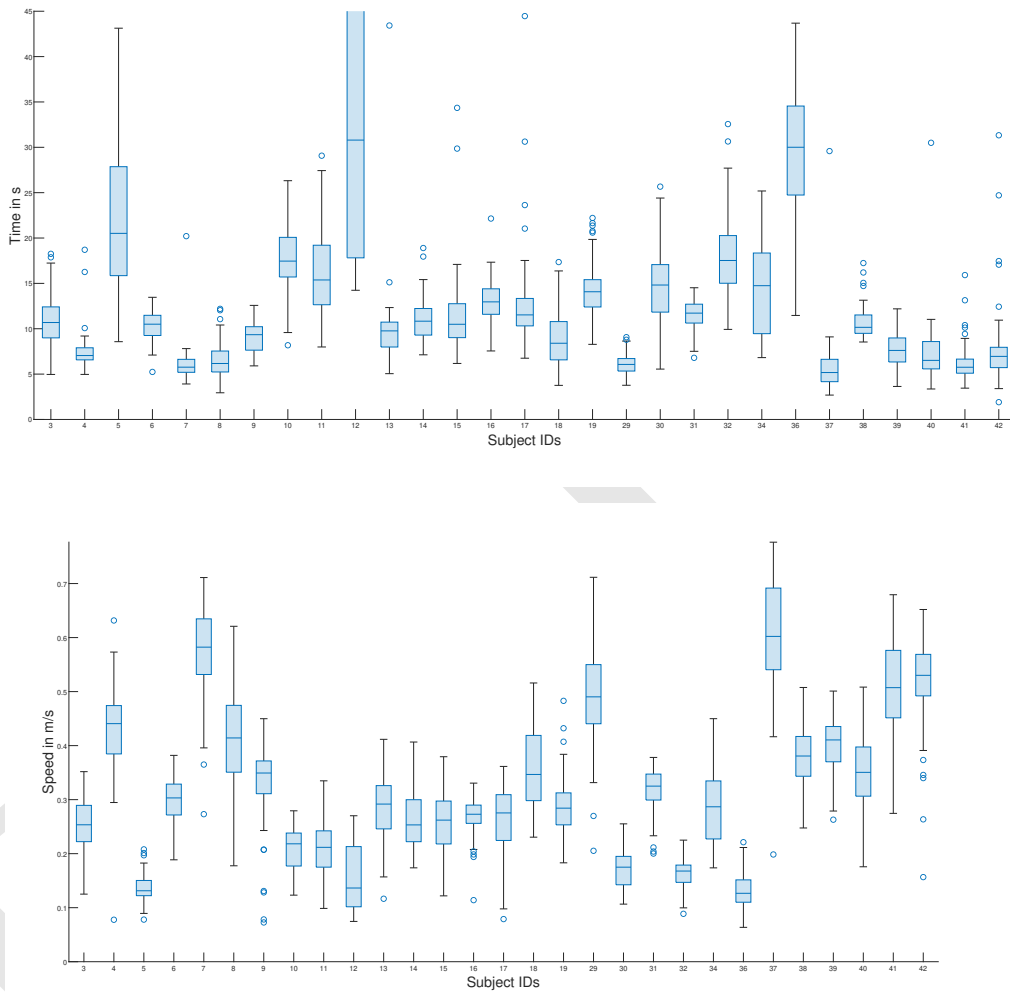


Fig. 4: Overview of the average trial completion time (top) and average walking speed (bottom) for all 72 trials grouped by subjects. Outliers outside the mean plus 3*std are excluded.

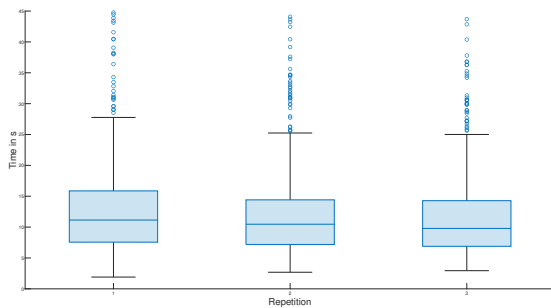


Fig. 5: Overview of the average walking speed for all 72 trials grouped ascending by the Repetition number. Outliers outside the mean plus 3*std are excluded.

was calculated considering the audio playback start as the starting point and the end point when the last click was provided. Fig.5 showcases the completion time grouped by the repetition number. Here, the time taken decreases with the presentation number. Fig. 4 (top) depicts how long different subjects took to complete the conditions. As expected, a variation can be observed. For the walking speed, the same time frame was considered. Fig. 4 (bottom) showcases the speed variation between the subjects. For a more detailed observation, the speed variation along L3 for four subjects can be observed in Fig. 6. In total 33 subjects participated in the test, out of which 18 are considered experienced. In Fig. 7 the attribute and confidence rating variation can be observed for the two subject groups.

2 Attribute and Click Evaluation

The following section includes an overview of the attribute and confidence ratings. Fig. 8,9, 10 and 11 depict the ratings of all subjects grouped by the dominant factor Line. ANOVA results are presented in Tables 1, 4. Additionally, art-con results are included in Tables 2,3,5, 6, 7, 8.

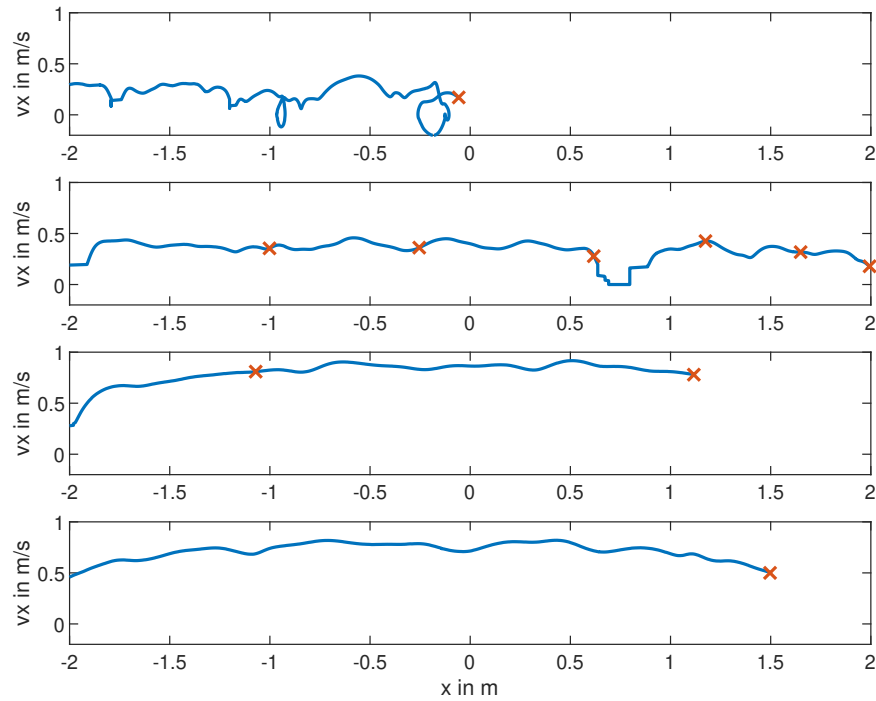


Fig. 6: Walking speed variation for four subjects for L3, S_2 , Repetition 3, WN, n2o. The red x indicates where the subjects clicked.

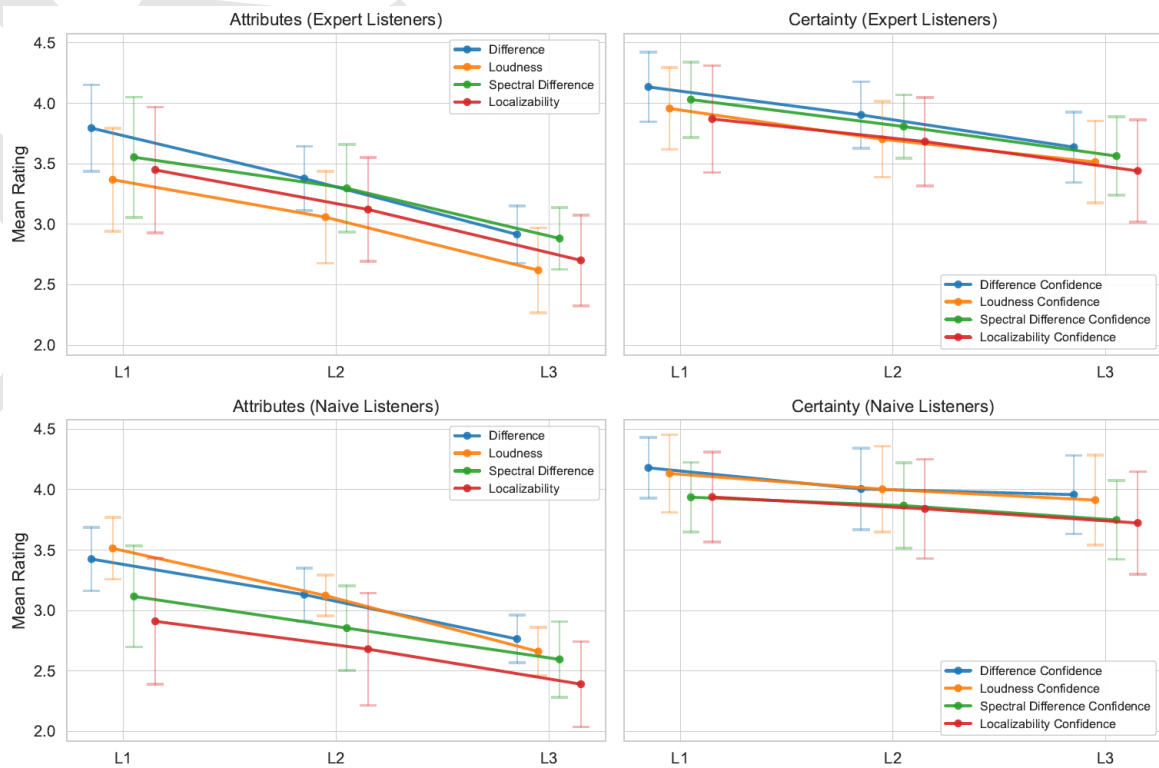


Fig. 7: Mean rating variation for the attributes and certainty for experienced and naive listeners.

The difference between the mean X-coordinate click values for the mean mode and raw multimodal distribution against the geometric boundary are also provided in Tables 9, 10, 11,12.

Table 1: ART ANOVA results for first and second order interactions considering the click distribution (**left**) and click confidence ratings (**right**). Statistically significant results are marked in bold.

Factor	p-value _{click}	p-value _{confidence}
Audio	< 0.001	< 0.001
Source	< 0.001	< 0.001
Direction	< 0.001	0.005
Line	< 0.001	< 0.001
Repetition	0.95	0.36
Audio × Source	0.91	0.13
Audio × Direction	< 0.001	< 0.001
Source × Direction	0.14	0.07
Audio × Line	0.94	0.001
Source × Line	< 0.001	< 0.001
Direction × Line	< 0.001	0.01
Audio × Repetition	0.78	0.83
Source × Repetition	0.997	0.42
Direction × Repetition	0.39	0.19
Line × Repetition	0.92	0.2

Table 2: Art-con results for the significant within single factor interactions for the considered factors and their impact on the click rating.

Factor	Estimate	SE	df	z.ratio	p-value
L3-L2	-252	47.9	Inf	-5.269	< .0001
L3-L1	-915	48.0	Inf	-19.071	< .0001
L2-L1	-663	47.3	Inf	-14.002	< .0001
$S_1 - S_2$	-156	40.4	Inf	-3.850	0.0001
S-WN	150	40.5	Inf	3.698	0.0002
n2o-o2n	1170	36.8	Inf	31.835	< .0001

Table 3: Art-con results for the significant within single factor interactions for the considered factors and their impact on the click **certainty**.

Factor	Estimate	SE	df	t.ratio	p-value
L3-L2	-245	27.3	2227	-8.99	< .0001
L3-L1	-463	27.3	2227	-16.957	< .0001
L2-L1	-218	27.2	2227	-7.986	< .0001
$S_1 - S_2$	-130	23.2	2227	-5.605	< .0001
S-WN	-201	23.1	2227	-8.701	< .0001
n2o-o2n	-65.5	23.3	2227	-2.812	0.005

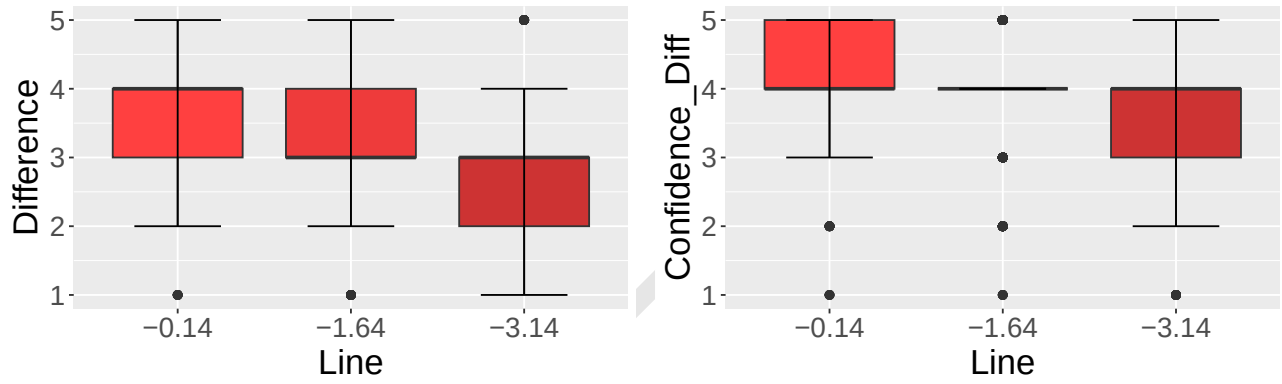


Fig. 8: General Difference ratings grouped by line (left). Corresponding confidence ratings grouped by line (right).

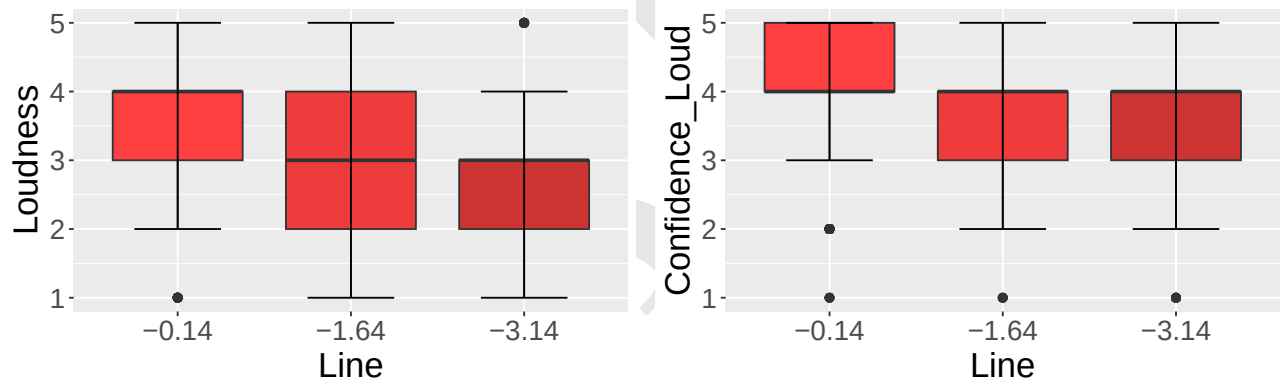


Fig. 9: Loudness ratings grouped by line (left). Corresponding confidence ratings grouped by line (right).

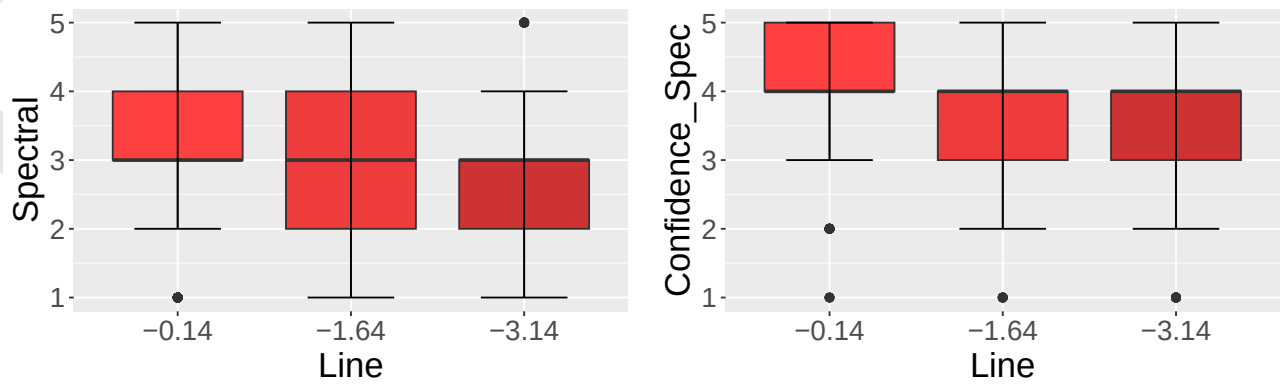


Fig. 10: Spectrum ratings grouped by line (left). Corresponding confidence ratings grouped by line (right).

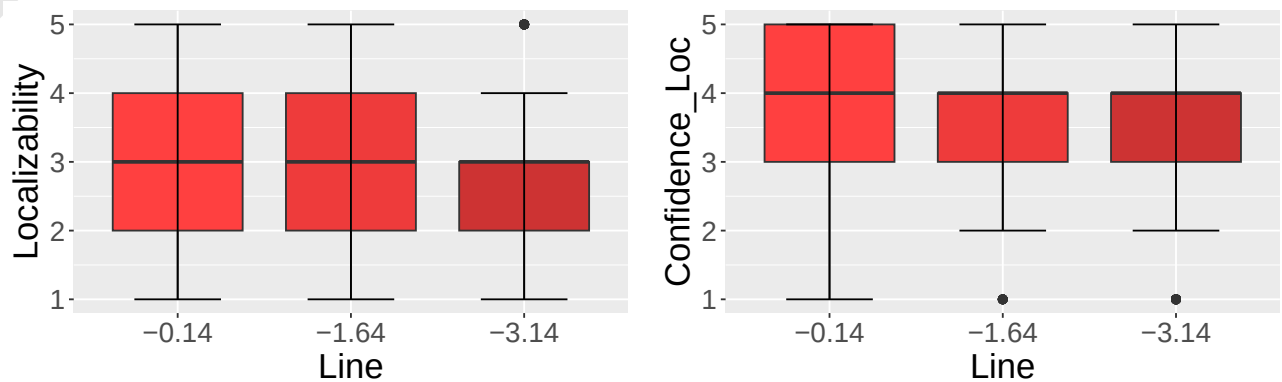


Fig. 11: Localizability ratings grouped by line (left). Corresponding confidence ratings grouped by line (right).

Factor	P_{diff}	$P_{diffconf}$	P_{loud}	$P_{loudconf}$	P_{spec}	$P_{specconf}$	P_{loc}	$P_{locconf}$
Audio	< 0.001	< 0.001	< 0.001	0.005	< 0.001	< 0.001	0.58	0.37
Source	< 0.001	0.05	< 0.001	0.04	0.6	0.14	0.01	0.15
Direction	0.69	0.16	0.9	< 0.001	0.17	0.46	0.93	0.14
Line	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Repetition	0.33	0.02	0.1	0.04	0.003	0.1	0.05	0.2
Audio $\times$ Source	0.04	0.36	0.79	0.89	0.89	0.47	0.13	0.66
Audio $\times$ Direction	< 0.001	< 0.001	< 0.001	0.002	0.001	0.14	0.27	0.61
Source $\times$ Direction	0.28	0.43	0.89	0.32	0.06	0.33	0.11	0.69
Audio $\times$ Line	0.02	0.41	0.14	0.002	0.13	0.42	0.27	0.11
Source $\times$ Line	< 0.001	< 0.001	< 0.001	0.01	< 0.001	0.09	< 0.001	0.02
Direction $\times$ Line	0.47	0.01	0.03	0.007	0.7	0.06	0.1	0.28
Audio $\times$ Repetition	0.12	0.29	0.27	0.11	0.03	0.21	0.03	0.03
Source $\times$ Repetition	0.99	0.01	0.1	0.99	0.5	0.13	0.88	0.79
Direction $\times$ Repetition	0.77	0.69	0.03	0.8	0.03	0.21	0.46	0.16
Line $\times$ Repetition	0.91	0.29	0.002	0.15	0.81	0.31	0.14	0.83

Table 4: ART ANOVA results for all the attributes and corresponding confidence ratings. Statistically significant results are marked in bold.

Factor	Estimate	SE	df	t.ratio	p-value
S, S_1 - S, S_2	-88.8	32.6	2227	-2.721	0.0332
S, S_1 - WN, S_1	-141.7	32.6	2227	-4.348	0.0001
S, S_1 - WN, S_2	-323.6	32.6	2227	-9.911	< .0001
S, S_2 - WN, S_1	-53.0	32.5	2227	-1.632	0.3608
S, S_2 - WN, S_2	-234.8	32.5	2227	-7.226	< .0001
WN, S_1 - WN, S_2	-181.8	32.5	2227	-5.601	< .0001
S,L1 - S,L2	265.86	38.0	2227	6.988	< .0001
S,L1 - S,L3	431.46	38.2	2227	11.293	< .0001
S,L1 - WN,L1	-128.47	38.0	2227	-3.379	0.0096
S,L1 - WN,L2	-9.24	38.0	2227	-0.243	0.9999
S,L1 - WN,L3	322.90	38.0	2227	8.498	< .0001
S,L2 - S,L3	165.60	38.2	2227	4.339	0.0002
S,L2 - WN,L1	-394.33	38.0	2227	-10.379	< .0001
S,L2 - WN,L2	-275.10	38.0	2227	-7.240	< .0001
S,L2 - WN,L3	57.04	38.0	2227	1.502	0.6628
S,L3 - WN,L1	-559.93	38.2	2227	-14.677	< .0001
S,L3 - WN,L2	-440.71	38.2	2227	-11.550	< .0001
S,L3 - WN,L3	-108.56	38.1	2227	-2.848	0.0506
WN,L1 - WN,L2	119.23	38.0	2227	3.140	0.0212
WN,L1 - WN,L3	451.37	37.9	2227	11.896	< .0001
WN,L2 - WN,L3	332.14	37.9	2227	8.754	< .0001
n2o,L1 - n2o,L2	264.2	38.6	2227	6.841	< .0001
n2o,L1 - n2o,L3	453.0	38.7	2227	11.706	< .0001
n2o,L1 - o2n,L1	17.1	38.6	2227	0.442	0.9979
n2o,L1 - o2n,L2	156.4	38.6	2227	4.046	0.0008
n2o,L1 - o2n,L3	465.4	38.7	2227	12.024	< .0001
n2o,L2 - n2o,L3	188.8	38.6	2227	4.888	< .0001
n2o,L2 - o2n,L1	-247.2	38.5	2227	-6.418	< .0001
n2o,L2 - o2n,L2	-107.8	38.6	2227	-2.796	0.0584
n2o,L2 - o2n,L3	201.2	38.6	2227	5.208	< .0001
n2o,L3 - o2n,L1	-435.9	38.6	2227	-11.296	< .0001
n2o,L3 - o2n,L2	-296.6	38.6	2227	-7.676	< .0001
n2o,L3 - o2n,L3	12.4	38.7	2227	0.321	0.9996
o2n,L1 - o2n,L2	139.3	38.5	2227	3.615	0.0042
o2n,L1 - o2n,L3	448.3	38.6	2227	11.615	< .0001
o2n,L2 - o2n,L3	309.0	38.6	2227	7.996	< .0001
S,n2o - S,o2n	-121.2	32.7	2227	-3.710	0.0012
S,n2o - WN,n2o	-280.1	32.6	2227	-8.591	< .0001
S,n2o - WN,o2n	-185.4	32.6	2227	-5.691	< .0001
S,o2n - WN,n2o	-158.9	32.6	2227	-4.879	< .0001
S,o2n - WN,o2n	-64.3	32.6	2227	-1.975	0.1979
WN,n2o - WN,o2n	94.6	32.5	2227	2.912	0.0190

Table 5: Art-con results for high-level significant factor interactions for the **clicking certainty**.

Factor	Estimate	SE	df	t.ratio	p-value
$S_{1,L1} - S_{1,L2}$	240.21	38.1	2227	6.304	< .0001
$S_{1,L1} - S_{1,L3}$	561.61	38.3	2227	14.674	< .0001
$S_{1,L1} - S_{2,L1}$	2.99	38.1	2227	0.078	1.0000
$S_{1,L1} - S_{2,L2}$	229.43	38.2	2227	6.010	< .0001
$S_{1,L1} - S_{2,L3}$	381.38	38.2	2227	9.997	< .0001
$S_{1,L2} - S_{1,L3}$	321.41	38.1	2227	8.429	< .0001
$S_{1,L2} - S_{2,L1}$	-237.22	37.9	2227	-6.251	< .0001
$S_{1,L2} - S_{2,L2}$	-10.78	38.0	2227	-0.283	0.9998
$S_{1,L2} - S_{2,L3}$	141.18	38.0	2227	3.713	0.0029
$S_{1,L3} - S_{2,L1}$	-558.63	38.1	2227	-14.658	< .0001
$S_{1,L3} - S_{2,L2}$	-332.19	38.2	2227	-8.694	< .0001
$S_{1,L3} - S_{2,L3}$	-180.23	38.2	2227	-4.720	< .0001
$S_{2,L1} - S_{2,L2}$	226.44	38.0	2227	5.956	< .0001
$S_{2,L1} - S_{2,L3}$	378.40	38.0	2227	9.959	< .0001
$S_{2,L2} - S_{2,L3}$	151.96	38.1	2227	3.989	0.0010
$S_{1,n2o} - S_{1,o2n}$	-4.35	32.9	2227	-0.132	0.9992
$S_{1,n2o} - S_{2,n2o}$	-100.19	32.9	2227	-3.050	0.0124
$S_{1,n2o} - S_{2,o2n}$	-172.09	32.8	2227	-5.245	< .0001
$S_{1,o2n} - S_{2,n2o}$	-95.83	32.9	2227	-2.917	0.0187
$S_{1,o2n} - S_{2,o2n}$	-167.73	32.8	2227	-5.111	< .0001
$S_{2,n2o} - S_{2,o2n}$	-71.90	32.8	2227	-2.193	0.1254

Table 6: Art-con results for high-level significant factor interactions for the **clicking certainty** continued.

Attribute-Factor	Estimate	SE	df	t.ratio	p-value
Difference -L3-L2	-304	27.8	2227	-10.943	< .0001
Difference -L3-L1	-552	27.8	2227	-19.878	< .0001
Difference -L2-L1	-248	27.7	2227	-8.957	< .0001
Loudness -L3-L2	-268	27.2	2227	-9.866	< .0001
Loudness -L3-L1	-502	27.2	2227	-18.455	< .0001
Loudness-L2-L1	-234	27.1	2227	-8.610	< .0001
Spectral -L3-L2	-222	26.4	2227	-8.412	< .0001
Spectral-L3-L1	-374	26.4	2227	-14.137	< .0001
Spectral-L2-L1	-151	26.4	2227	-5.740	< .0001
Localizability -L3-L2	-224	25.1	2227	-8.925	< .0001
Localizability -L3-L1	-371	25.1	2227	-14.748	< .0001
Localizability -L2-L1	-147	25.1	2227	-5.839	< .0001
Difference -S ₁ -S ₂	-84	23.9	2227	-3.522	0.0004
Loudness -S ₁ -S ₂	-89.6	23.2	2227	-3.866	0.0001
Spectral -S ₁ -S ₂	-11.5	22	2227	-0.521	0.6022
Localizability -S ₁ -S ₂	-54.2	21.2	2227	-2.564	0.0104
Difference-S-WN	-227	23.6	2227	-9.619	< .0001
Loudness-S-WN	-143	23.1	2227	-6.178	< .0001
Spectral-S-WN	-307	21.5	2227	-14.301	< .0001
Localizability-S-WN	11.7	21.2	2227	0.551	0.5815
Difference-n2o-o2n	9.55	23.9	2227	0.4	0.6892
Loudness-n2o-o2n	-2.83	23.2	2227	-0.122	0.903
Spectral-n2o-o2n	30	22	2227	1.361	0.1738
Localizability-n2o-o2n	-1.86	21.2	2227	-0.088	0.9299
Difference -Rep1 - Rep2	42.9	29.2	2227	1.471	0.305
Difference -Rep1 - Rep3	27.7	29.2	2227	0.947	0.6105
Difference- Rep2 - Rep3	-15.2	29.3	2227	-0.52	0.2227
Loudness -Rep1 - Rep2	48.82	28.4	2227	1.721	0.1975
Loudness -Rep1 - Rep3	56.55	28.4	2227	1.989	0.1150
Loudness- Rep2 - Rep3	7.74	28.5	2227	0.272	0.9601
Spectral -Rep1 - Rep2	62.8	26.9	2227	2.338	0.0510
Spectral -Rep1 - Rep3	89.8	26.9	2227	3.333	0.0025
Spectral -Rep2 - Rep3	26.9	27.0	2227	0.998	0.5781
Localizability-Rep1 - Rep2	44.6	25.8	2227	1.727	0.1954
Localizability -Rep1 - Rep3	61.0	25.8	2227	2.357	0.0485
Localizability -Rep2 - Rep3	16.4	25.9	2227	0.633	0.802

Table 7: Art-con results for the significant within-level interactions for the considered **attributes**. Bold values indicate the presence of insignificant group comparisons.

Attribute	Estimate	SE	df	t.ratio	p-value
Difference -L3-L2	-111	25.3	2227	-4.379	< .0001
Difference -L3-L1	-296	25.3	2227	-11.683	< .0001
Difference -L2-L1	-185	25.3	2227	-7.320	< .0001
Loudness -L3-L2	-64.4	23.9	2227	-2.700	0.0191
Loudness -L3-L1	-260.1	23.9	2227	-10.896	< .0001
Loudness-L2-L1	-195.7	23.8	2227	-8.213	< .0001
Spectral -L3-L2	-128	26.4	2227	-4.835	< .0001
Spectral-L3-L1	-247	26.4	2227	-9.334	< .0001
Spectral-L2-L1	-119	26.4	2227	-4.510	< .0001
Localizability -L3-L2	-111	23.8	2227	-4.676	< .0001
Localizability -L3-L1	-212	23.8	2227	-8.921	< .0001
Localizability -L2-L1	-101	23.7	2227	-4.255	0.0001
Difference -S ₁ -S ₂	-41.2	21.2	2227	-1.943	0.0521
Loudness -S ₁ -S ₂	-40.5	19.9	2227	-2.029	0.0425
Spectral -S ₁ -S ₂	-32.7	21.9	2227	-1.492	0.1359
Localizability - S ₁ -S ₂	-28.2	19.7	2227	-1.434	0.1518
Difference - S-WN	-134	21	2227	-6.388	< .0001
Loudness - S-WN	-55.9	19.9	2227	-2.806	0.0051
Spectral - S-WN	-145	21.7	2227	-6.662	< .0001
Localizability - S-WN	-17.7	19.7	2227	-0.899	0.3689
Difference-n2o-o2n	-29.5	21.2	2227	-1.394	0.1634
Loudness-n2o-o2n	-72.8	19.9	2227	-3.654	0.0003
Spectral-n2o-o2n	16.1	21.9	2227	0.736	0.1738
Localizability-n2o-o2n	-1.86	21.2	2227	-0.088	0.4619
Difference - Rep1 - Rep2	72	25.9	2227	2.778	0.0152
Difference - Rep1 - Rep3	41.9	26.0	2227	1.616	0.2391
Difference - Rep2 - Rep3	-30	26.0	2227	-1.154	0.4807
Loudness -Rep1 - Rep2	26.9	24.4	2227	1.104	0.5117
Loudness -Rep1 - Rep3	62.4	24.4	2227	2.553	0.029
Loudness- Rep2 - Rep3	35.5	24.5	2227	1.449	0.3162
Spectral -Rep1 - Rep2	47.4	26.7	2227	1.773	0.179
Spectral -Rep1 - Rep3	51.68	26.8	2227	1.929	0.1309
Spectral -Rep2 - Rep3	4.27	26.8	2227	0.159	0.9861
Localizability-Rep1 - Rep2	35.3	24.0	2227	1.469	0.3063
Localizability -Rep1 - Rep3	39.59	24.1	2227	1.64	0.2276
Localizability -Rep2 - Rep3	4.29	24.1	2227	0.178	0.9827

Table 8: Art-con results for the significant within-level interactions for the considered **attribute certainty** ratings. Bold values indicate the presence of insignificant group comparisons

	S1		S2	
	n2o	o2n	n2o	o2n
L1	xp = 1.23 δ = 0.55	xp = 0.4 δ = 0.28	xp = 1.26 δ = 0.43	xp1 = -0.31; xp2 = 0.55 δ 1 = 1.14; δ 2 = 0.28
L2	xp = 0.51 δ = 0.89	xp = -0.59 δ = 0.21	xp = 0.83 δ = 0.61	xp = -0.41 δ = 0.63
L3	xp = -0.42 δ = 1.02	xp = 0.04 δ = 1.48	xp = 0.48 δ = 0.86	xp = -1.07 δ = 0.69

Table 9: X-coordinate (xp) **main mode** peak/mean value and the difference (δ) in meters to the acoustic boundary for the **speech sample**.

	S1		S2	
	n2o	o2n	n2o	o2n
L1	xp = 0.70 δ = 0.03	xp = 0.07 δ = 0.61	xp = 0.77 δ = 0.05	xp = 0.09 δ = 0.73
L2	xp = 0.50 δ = 0.89	xp = -0.58 δ = 0.21	xp = 0.64 δ = 0.43	xp = -0.40 δ = 0.63
L3	xp = 0.24 δ = 1.69	xp = -0.67 δ = 0.76	xp = 0.47 δ = 0.86	xp = -0.64 δ = 0.26

Table 10: Mean X-coordinate (xp) value from the **raw multimodal** distribution and the difference (δ) in meters to the acoustic boundary for the **speech sample**.

	S1		S2	
	n2o	o2n	n2o	o2n
L1	xp = 1.16 δ = 0.48	xp = 0.03 δ = 0.65	xp = 1.25 δ = 0.42	xp1 = 0.08; xp2 = 0.53 δ 1 = 0.75; δ 2 = 0.3
L2	xp = 0.22 δ = 0.6	xp = -0.96 δ = 0.58	xp = 0.72 δ = 0.5	xp = -0.32 δ = 0.54
L3	xp = -0.87 δ = 0.57	xp = 0.0 δ = 1.44	xp = 0.21 δ = 0.59	xp = -0.95 δ = 0.57

Table 11: X-coordinate (xp) **main mode** peak/mean value and the difference (δ) in meters to the acoustic boundary for the **noise sample**.

	S1		S2	
	n2o	o2n	n2o	o2n
L1	xp = 0.63 δ = 0.04	xp = -0.03 δ = 0.72	xp = 0.62 δ = 0.21	xp = 0.02 δ = 0.8
L2	xp = 0.22 δ = 0.6	xp = -0.57 δ = 0.19	xp = 0.43 δ = 0.22	xp = -0.31 δ = 0.54
L3	xp = 0.04 δ = 1.49	xp = -0.56 δ = 0.88	xp = 0.20 δ = 0.59	xp = -0.59 δ = 0.21

Table 12: Mean X-coordinate (xp) value from the **raw multimodal** distribution and the difference (δ) in meters to the acoustic boundary for the **noise sample**.